

A Techno-Economic Modelling and Component Sizing in Renewable Energy Communities: The Perspective of Technical Facilitators

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Abstract—This article presents a mathematical model to find the optimal size of solar Photovoltaics (PVs) and Battery Energy Storage Systems (BESSs) within the distinctive framework of a Renewable Energy Community (REC) with energy exchanges among PV+BESS system and participants. Our optimization considers dynamic factors, such as electricity pricing and financial aspects including capital investment and operational costs of PV-BESS systems. The replacement costs and efficiencies associated with charging and discharging BESS units are also considered. Mixed Integer Non-Linear Programming (MINLP) is employed to maximize the Net Profit (NP) of the REC. Additionally, we introduce an individual marginal contribution method to allocate incentives among REC participants equitably. This article provides practical insights for Technical Facilitators (TFs) to guide them in selecting the most suitable system size for their energy requirements in the REC. To show real-world applicability, we present the case study of a REC in Roseto Valfortore, a small town in the Province of Foggia, Italy.

Index Terms—Optimal sizing, Solar Photovoltaics, Battery Energy Storage Systems, Renewable Energy Community

I. INTRODUCTION

A Renewable Energy Community (REC) often involves consumers and renewable energy sources owned by the Technical Facilitators (TF) to collectively benefit from Public Authority (PA) incentives. In one of the typical scenarios, the TF signs a contract for a Rooftop Lease Agreement (RLA) with a private company to install solar Photovoltaics (PVs) and Battery Energy Storage Systems (BESSs). In this setup, TF's lease payments are based on contractual terms, eliminating the need for additional contracts with REC participants for rooftop space. This model allows TF to decide how PA incentives should be distributed among participants. For the effective operation of this model, communication between TF and PA is crucial to get incentives on energy sharing [1], [2]. Also, participant aggregation reduces the burden on the power network, prompting extra incentives from authorities and revenue generation by selling power to the grid [3], [4]. Apart from this, REC brings a triple benefit: reduced carbon footprint, enhanced social participation, and local energy security during grid failure. To maximize the profit

for the entire REC, predetermined sizes for PVs and BESSs are required. Prior studies have addressed sizing challenges without explicitly considering revenue from energy sharing or potential savings on transmission costs. For instance, in [5], we see optimal sizing with solar PVs, BESSs, and diesel generators. In [6], a sizing optimization problem for solar PVs and BESSs is formulated using real-world data from 300 Australian households without calculating the systems' Net Present Value (NPV). The authors in [7] and [8] introduced a model to size solar PVs and BESSs in residential systems, accounting for uncertainties in solar generation and load demand without considering the NPV. Solar PVs and BESSs are used in various areas such as peer-to-peer energy trading [9], demand-side management [10], and reducing electricity bills [11]. Key performance indicators, with component sizing, are outlined in [12]. However, more research is needed to focus on the optimal sizing of PVs and BESSs for the entire REC. For instance, [13] suggests a model maximizing community revenue, while [14] introduces fair revenue sharing among REC participants based on optimal sizing of PVs and BESSs. The authors in [15] elaborate on revenue distribution within REC, considering internal local market dynamics. Notably, [16] and [17] proposed linear programming for optimal PVs and BESSs sizing. Moreover, [18] uses a bi-level optimization model to determine the parameters and profiles of new REC participants. In energy trading, [19] explores power purchase and bidding strategies for REC communities.

While the mentioned articles have made progress in tackling the sizing problem, they still need to consider the NPV of the systems, which is arguably a crucial factor. Equally important is that many studies focus only on single-day profiles, overlooking the critical aspect of an extended planning horizon (typically 25 years). This article uses realistic household demand and solar generation data with seasonal fluctuations to optimize the size of the system by exploiting inherent periodicity in load demand and solar generation profiles. In addition to this, our research introduces the individual marginal contribution as a computationally efficient mechanism for distributing incentives. We thoroughly examined this new approach both analytically and numerically, uncovering insights into its properties and effectiveness within the REC. This article aims to fill essential gaps in current research on the optimal sizing of PVs and BESSs in the context of the REC.

The remainder of the paper is structured as follows: Section II describes the mathematical model of the system components; in Section III, we formulate the optimal sizing problem followed by its approximation using “periodic

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The TF also pays for the operating and maintenance expenses of the BESS (OPEX^b), alongside the rooftop lease agreement (RLA^b) costs. The combined annual cost is defined as

$$\text{OMLA}_y^b = N^b \text{OPEX}_y^b + N^b \text{RLA}_y^b. \quad (13)$$

C. Power flows

The power flows established in the case under consideration are depicted in Fig. 1, where i) P_t^c denotes the charging of the BESSs by the generated solar power; ii) P_t^{sell} denotes the left-over solar power (after charging the BESS) injected into the grid; iii) P_t^{sh} denotes the shared power, i.e., the portion of the aggregate community demand which is met using P_t^{sell} , *instant by instant*; iv) P_t^d denotes the discharging of the BESSs, which happens only when the aggregate demand cannot be met using the shared power; v) P_t^{buy} denotes the left over demand which cannot be met using the shared power and the BESS, rather is purchased from the grid. The above power flows are modeled as the following load balance constraints:

$$P_t^{\text{sell}} = P_t^s - P_t^c, \quad (14a)$$

$$P_t^{\text{sh}} = \min \left(P_t^{\text{sell}}, \sum_{n \in \mathcal{N}} d_t^n \right), \quad (14b)$$

$$P_t^{\text{buy}} = \sum_{n \in \mathcal{N}} d_t^n - P_t^{\text{sh}} - P_t^d, \quad (14c)$$

where $0 \leq P_t^c \leq P_t^s$, d_t^n is the electricity demand of the n^{th} participant at time t , and $\mathcal{N}$ is a set of participants.

D. Economic exchanges

The TF and REC participants are associated with economic exchanges owing to different power flows (see II-C). Let $C_t^S \in \mathbb{R}^+$ denote the tariff rate at which unit power is sold to the grid and let $C_t^E \in \mathbb{R}^+$ and $C_t^T \in \mathbb{R}^+$, respectively denote, the incentive on unit power shared within the REC and the incentive on unit saving in transmission power due to sharing. The TF is compensated by the PA for the power injected into the grid and for the power shared among the community members, respectively defined as

$$R_y^{\text{sell}} = \sum_{t \in \mathcal{T}_y} C_t^S P_t^{\text{sell}} \Delta t \quad (15)$$

and

$$I_y^{\text{sh}} = \sum_{t \in \mathcal{T}_y} \underbrace{(C_t^E + C_t^T) P_t^{\text{sh}} \Delta t}_{:= I_{y,t}^{\text{sh}}}. \quad (16)$$

Likewise, let $C_t^P \in \mathbb{R}^+$ denote the tariff rate at which unit power is purchased from the grid, wherein $C_t^P \geq C_t^S$ for presumably all real-world scenarios. The REC participants are collectively billed by the PA for the net energy purchased from the grid, defined as

$$C_y^{\text{buy}} = \sum_{t \in \mathcal{T}_y} C_t^P P_t^{\text{buy}} \Delta t. \quad (17)$$

As we assumed the community shared BESS is physically connected with the REC participants, it can be discharged to meet the electricity demand (denoted as d_t^n , see II-C). This

results in an economic saving, as otherwise, the same amount of electricity would have been purchased from the grid at a higher price, i.e., C_t^P . We model this as a form of saving for the REC participants, defined as

$$R_y^{\text{sav}} = \sum_{t \in \mathcal{T}_y} C_t^P P_t^d \Delta t. \quad (18)$$

E. NPV

The NPV is the present value of cash flows (CF) at a given rate of return [22]. The BESSs are expected to be replaced at least once within the typical time horizons related to the optimization problem proposed in the next section. In this work, we considered the lifespan of the BESSs to be L^b years, after which it needs to be replaced. Using the purchasing cost of N^b batteries as per (12), we compute the present value for BESSs as

$$C^b = \underbrace{\hat{C}^b N^b \left[\frac{1 + (1+r)^{L^b}}{(1+r)^{L^b}} \right]}_{\text{Total investment for BESS}}, \quad (19)$$

where r is the discount rate. The cumulative computation of NPV is obtained by adding up each year's CFs over the entire planning horizon and hence, from (5), (13), (15), (16) and (18) it is given by

$$\text{NPV}_m = \sum_{y=0}^m \frac{-\text{INV}_y + R_y^{\text{sell}} + R_y^{\text{sav}} + \delta I_y^{\text{sh}} - C_y^{\text{buy}} - \text{OMLA}_y^p - \text{OMLA}_y^b}{(1+r)^y}, \quad (20)$$

where δ is a percentage of the total incentive the TF wants to keep, and INV_y denotes the investment by the TF corresponding to y^{th} year, defined as

$$\text{INV}_y = \begin{cases} C^p + \bar{C}^b, & y = 0 \text{ (initial investment),} \\ \bar{C}^b, & y = L^b \text{ (BESS is replaced),} \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

From the Eqn. 20, the payback period (PB) is calculated as follows:

$$\text{PB} = \min \left\{ m \mid \text{NPV}_m \geq 0 \right\}. \quad (22)$$

The payback period is often used as a simple way to evaluate the profitability and risk of an investment. After the payback period, the investment continues to generate cash flows for the remainder of the lifespan of PV-BESS systems, which is considered as a total return. Now, the Net Profit (NP) for the entire REC is calculated as

$$\text{NP} = \text{NPV}_{|\mathcal{Y}|}, \quad (23)$$

where the cardinality of $\mathcal{Y}$, denoted as $|\mathcal{Y}|$, is the length of the planning horizon, which includes all the periods considered for the investment evaluation. The NP thus comprises all future cash flows discounted back to their present value, providing a measure of the financial performance of the REC investment over the lifespan.

III. OPTIMAL SIZING PROBLEM FORMULATION

An optimization problem is formulated for the considered REC to find an optimal number of solar PVs and BESSs while maximizing the NP of the entire system. Formally,

$$\begin{aligned} \max_{\mathcal{X}} \quad & \underbrace{(20)}_{\text{Net Profit}} \\ \text{s.t.} \quad & \underbrace{(14a) \text{ to } (14c)}_{\text{Power balance constraints}} \\ & \underbrace{(3), (8a) \text{ to } (11)}_{\text{Operational constraints}} \end{aligned} \quad (24)$$

where $\mathcal{X} = \{N^p, N^b, P_{t,y}^{\text{buy}}, P_{t,y}^{\text{sell}}, P_{t,y}^{\text{sh}}, P_{t,y}^c, P_{t,y}^d, b_t^c, b_t^d, S_{t,y} \mid \forall t \in \mathcal{T}_y, \forall y \in \mathcal{Y}\}$ is the set of decision variables and $\mathcal{T}_y := \{1, 2, \dots, T_y\}$, $\forall y \in \mathcal{Y}$ is the planning horizon. The yearly horizon is typically the same for all years, i.e., $T_y = \bar{T}$, $\forall y \in \mathcal{Y}$. We seek to maximize the NP by finding the optimal value of each variable that produces the highest output value. Solving (24) for a planning horizon of the order of decades is computationally very heavy due to the increase in the number of decision variables and constraints to be satisfied at each time instant. As an alternative to it, we propose exploiting the approximate periodicity of solar generation and the demand profiles, which are described in the subsequent subsection.

A. Exploiting Periodicity

We extracted weekly periodic profiles for spring, summer, fall, and winter from historical demand and solar generation data using six different methods, namely L1 norm, L2 norm, WLS, WLS + L1 regularization, average, and bound average methods explained in the article [21]. Among all these methods, we found the L1 norm method outperformed, and we used it to find weekly representative profiles of four seasons for solar generation and electricity demand of all REC participants. Using these extracted periodic profiles, (20) is recalculated as follows:

$$\overline{\text{NPV}}_m = \sum_{y=0}^m \frac{-\text{INV}_y + \beta(R_x^{\text{sell}} + R_x^{\text{sav}} + \delta I_x^{\text{sh}} - C_x^{\text{buy}}) - \text{OMLA}_y^p - \text{OMLA}_y^b}{(1+r)^y}, \quad (25)$$

where R_x^{sell} , R_x^{sav} , I_x^{sh} , C_x^{buy} respectively are revenues for selling power to the grid, savings on self-consumption, incentive on sharing power and costs to buy power from the grid for four representative periodic seasonal weeks. To obtain the approximate annual revenues and costs, we multiplied it by $\beta = 13$, which is the total number of weeks over one season. From Eqn. (25), the NP of the entire REC is calculated using the expression as follows:

$$\overline{\text{NP}} = \overline{\text{NPV}}_{|\mathcal{Y}|}. \quad (26)$$

This updated $\overline{\text{NP}}$ obtained using periodic signals is further used in the optimization problem to make the most economically efficient configuration for the REC.

B. Optimal Sizing using Periodic Profiles

A new optimization problem is cast as follows:

$$\begin{aligned} \max_{\mathcal{X}} \quad & \underbrace{(26)}_{\text{Net Profit}} \\ \text{s.t.} \quad & \underbrace{(14a) \text{ to } (14c)}_{\text{Power balance constraints}} \\ & \underbrace{(3), (8a) \text{ to } (11)}_{\text{Operational constraints}} \cap \text{State of charge periodicity}^1 \end{aligned} \quad (27)$$

where $\hat{\mathcal{X}} = \{N^p, N^b, P_t^{\text{buy}}, P_t^{\text{sell}}, P_t^{\text{sh}}, P_t^c, P_t^d, b_t^c, b_t^d, S_t \mid \forall t \in \hat{\mathcal{T}}\}$ is set of decision variables and $\hat{\mathcal{T}} := \{1, 2, \dots, \hat{T}\}$ is set of hours over four periodic seasonal weeks with the total number of hours $\hat{T}$.

C. Individual Marginal Contribution

This method calculates the marginal contribution of each participant based on their individual energy consumption profile, which quantifies the unique contribution of each participant. For individual contribution, it is necessary to find the total incentive received from the PA and savings on self-consumption expressed as:

$$I_x^{\text{sh}} = \sum_{t \in \mathcal{T}_x} \underbrace{(C_t^E + C_t^T) P_t^{\text{sh}} \Delta t}_{:= I_{x,t}^{\text{sh}}}. \quad (28)$$

Let us assume that the TF retains $\delta\%$ of the total incentive. Hence, the revenue to be distributed among all participants is

$$R_{x,t}^{\text{sh}} = (1 - \delta) I_{x,t}^{\text{sh}}. \quad (29)$$

As per the individual participant's consumption, the distribution factor is expressed as

$$\zeta_{x,t}^n = \frac{d_t^n \Delta t}{\sum_{n \in \mathcal{N}} d_t^n \Delta t}, \quad (30)$$

where $\zeta_{x,t}^n$ is the proportion of total incentive received by the n^{th} participant proportional to their individual electricity consumption at time t . Now, the electricity bill of the n^{th} participant after joining the REC can be formulated as

$$EB^{\text{REC},n} = EB^{\text{IND},n} - \beta \sum_{t \in \hat{\mathcal{T}}} \zeta_{x,t}^n R_{x,t}^{\text{sh}}, \quad (31)$$

where $EB^{\text{IND},n}$ is annual electricity bill before joining the REC and $EB^{\text{REC},n}$ is annual electricity bill after joining the REC.

IV. SIMULATION RESULTS

The optimization algorithm is implemented using MATLAB software with the YALMIP toolbox [23] and Gurobi solver [24]. The hardware used for the simulation is an Intel Core i7-11800H, 2.3GHz server with 32GB of RAM. The developed model of the REC has been applied to the real case study of Roseto Valfortore. It is a small town

¹This state of charge periodicity can be imposed as $S_{w\hat{T}} = S_{(w-1)\hat{T}}$, where $w \in \{1, 2, 3, 4\}$ denotes the number of weeks and $\hat{T}$ denotes the number of samples in each week.

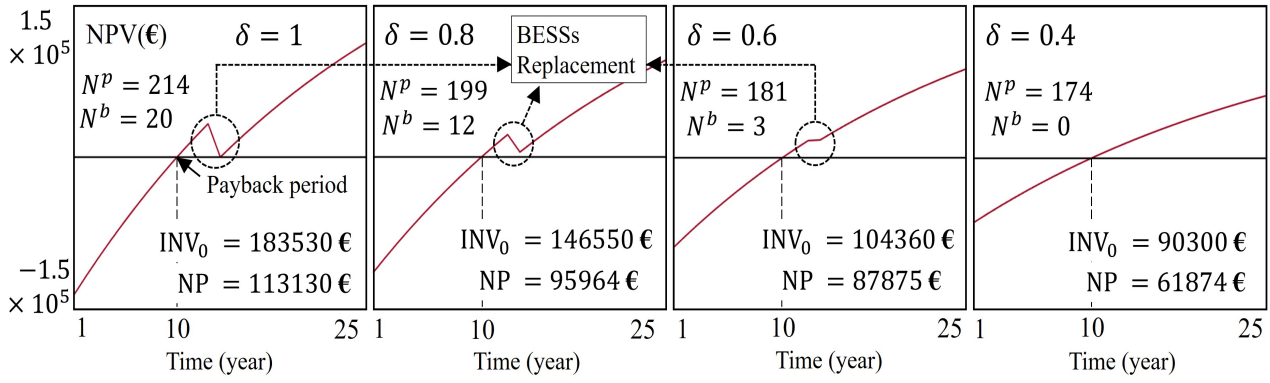


Fig. 2: The net present value (NPV), investment (INV), Net Profit (NP), and an optimal number of PVs & BESSs with four different values of δ .

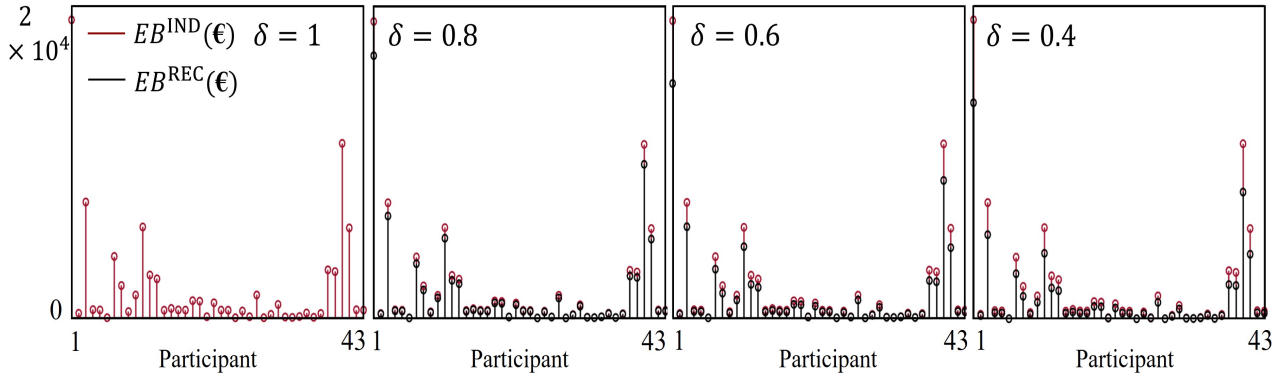


Fig. 3: Comparison of individual participants' electricity bills over the year, illustrating the impact of joining the renewable energy community (REC) versus remaining outside the community.

and municipality of the Province of Foggia in the Apulia Region of southeastern Italy. For the simulation analysis, all participant household demand profiles are collected from the TF, and historical solar irradiance and temperature data is collected from the online monitoring dashboard [25], with the sampling rate $\Delta t = 1$ hr. Furthermore, the solar PV panel and BESS specifications are mentioned in table I, which are taken from [27] and [26], respectively. Also, the buying and selling electricity rates are the same as in our previous article [21], whereas the sharing electricity rate C_t^E and electricity transmission rate C_t^T are 0.11 €/kWh and 0.00822 €/kWh, respectively.

The simulation results, illustrated in Fig. 2, involve forty-three participants with an annual total demand of 36.5 MWh and $\delta \in \{0.4, 0.6, 0.8, 1\}$. Notably, when $\delta = 1$ (i.e., TF keeps the entire incentives received from the PA), the optimal number of solar PVs and BESSs is higher than in the other three cases. Additionally, by imposing a constraint on the payback period, the TF can recover the investment cost within the specified period, aligning with the predetermined payback period for the REC. With the replacement of BESSs in the 12th year, there is a slight dip in NPV for $\delta = 0.6, 0.8, 1$, followed by a subsequent increase due to the replacement of BESSs. Moreover, when $\delta = 0.4$ (i.e., the

TABLE I: Specifications of the solar PV and BESS

Description	Short form	Value
PV		
Rating of PV-panel	P^r	0.43 kW
Size of one panel	A^p	1.95 m ²
PV-panel lifespan	L^p	25 years
Power temperature coefficient	γ	0.043 %/°C
Cost of PV	$\hat{C}^p$	1200 €/kW
Nominal operating cell temperature	θ^{nt}	45°C
Irradiance during STC	$\hat{E}$	1 kW/m ²
Temperature during STC	$\hat{\theta}$	25°C
Operating & maintenance costs	OPEX ^p	20€/kW/year
Rooftop lease agreement costs	RLA ^p	10€/kW/year
BESS		
Normalized capacity	B	5 kWh
Charging/Discharging efficiency	η^c, η^d	90%
Min Charging/Discharging power	$\underline{P}^c, \underline{P}^d$	0 kW
Max Charging/Discharging power	$\overline{P}^c, \overline{P}^d$	0.5 kW
Min state of charge	$\underline{S}$	0.5 kWh
Max state of charge	$\overline{S}$	& 4.5 kWh
Cost to buy BESS	$\hat{C}^b$	450 €/kWh
BESS lifespan	L^b	12 years
Operating & maintenance costs	OPEX ^b	24€/kWh/year
Rooftop lease agreement costs	RLA ^b	1€/kWh/year

TF retains 40% of the incentive and distributes the remaining 60% among all participants based on their individual shared energy consumption), the optimal quantities of solar PVs are

comparatively lower than in the other three scenarios, while the optimal number of BESSs is zero because of its higher investment cost to recover in the payback period.

Fig. 3 depicts a comparative analysis of annual electricity bills for all forty-three participants with and without joining the REC. When $\delta = 1$, indicating no distribution of incentives among participants, the electricity bills exhibit no significant difference between joining and not joining the REC. In contrast, for $\delta = 0.4$, where TF keeps 40% incentive received from the PA, and 60% of the total incentive is distributed among participants based on their individual contributions, the electricity bills after joining REC are nearly 30% lower compared to without joining the REC as expected. This observation indicates the attractiveness of REC participation, particularly as the value of δ decreases. The incentives for all REC participants depend on the value of δ , which is within the control of the TF. This effect of δ on energy transactions in the REC shows how important it is to balance participant profits and TF's efforts to recover their investment.

V. CONCLUSION

In this article, we introduced a novel model of REC, where TF invests in installing solar PVs and BESSs, and community members agree to receive part of the incentive. We framed the MINLP problem to find a suitable system size for the installations. Solving the MINLP for a longer planning horizon (25 years in our case) is computationally challenging. As an efficient approach, we used weekly representative profiles of demand and solar generation for each season of the year to approximate the optimization problem. This approach helps to efficiently determine the optimal number of PVs and BESSs needed for the REC. We also proposed a revenue-sharing mechanism based on individual contributions to energy sharing within the REC. Our simulations, involving 43 households, showed that the TF can recover its investment in the predefined payback period, and participants receive additional incentives for being part of the REC.

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